

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC HVDC Template

vhvdc1

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 Introduction

This document describes the usage of the WECC HVDC (vhvdc1) model implemented in *PowerFactory*.The model is based on [1]. The model is a simple planning level non-vendor specific VSC-HVDC dynamic model for point-to-point transmission in existing AC networks (not for offshore wind applications). It is a generic model to be used in high-level stability (balanced RMS) and power flow studies.

Chapters 2 and 3 describe how to use and configure the model in *PowerFactory*.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. The user has the task to adjust static and dynamic model parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific HVDC interconnection!

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2 Model Usage

The model template consists of two static generators (*ElmGenstat*) each connected to a terminal and a composite model which defines the dynamics. The composite model contains the dynamic model in DSL (*DIgSILENT* Simulation Language) and the measurement devices. Note that the simplified DC-link model is included in the DSL model. Therefore, it is not depicted in the single line diagram.

The models are available as templates and are accessible through the button *General Templates* in the drawing toolbox.

3 Model Configuration

The active power dispatch, dynamic model configuration and dynamic converter model are described in the following sections.

3.1 Active Power Dispatch

The template comprises one rectifier (modelled as a static generator that consumes active power) and one inverter (modelled as a static generator that supplies active power). Which converter takes on which role depends on the active power setting on the *Load Flow* page. The following QDSL (Quasi-Dynamic Simulation Language) algorithm is implemented to be used for power flow calculation and initialization of dynamic simulations: If the active power setpoint of converter 1 is larger than zero, converter 1 is the inverter and converter 2 is the rectifier. The active power consumption of the rectifier depends on the active power supplied by the inverter and the line and converter losses and is set accordingly. If the active power of converter 1 is smaller or equal to zero, the roles of the converters are reversed. The QDSL code is part of the QDSL model 'HVDC Link Load Flow for WECC VHVDC1' which is automatically added to the grid folder with the template.

Note also that the rated apparent power on the *Basic Data* page of the static generators should comply with the setting in the dynamic model described in 3.2.

3.2 Dynamic Model Configuration

The dynamic simulation is started from steady state, which is the load flow result. It is therefore important to check the load flow configuration of the elements.

The dynamic model can either be accessed via the generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the *Basic Data* page and click on the arrow button after "Model".
- Via Data Manager: Search the corresponding composite model 'HVDC Model', right click and select "Edit".

After opening the composite model the dynamic (common) model of the controller (including DC-link) can be accessed via a double click on the 2nd column.

The dynamic (DSL) model can be configured via scalar input parameters. These parameters are displayed after opening a common model on the *General* page.

3.3 Dynamic Converter Model

[1] does not specify the exact implementation of the dynamic converter model. The static generators in the template represent a current source model, which supplies current according to the active/reactive power setpoints. The dynamics of the current source model, i.e. its time constant, can be set on the *Simulation RMS* page of the static generator. Please refer to the technical reference of the static generator [2] for further details. Furthermore, the current controller has current limits which are set in the dynamic (DSL) model (parameters *I_lim_term1_pu/I_lim_term2_pu*). Note that these current limits are specific to the implementation of the template and are not included in the model specification by WECC [1].

4 References

[1] WECC - vHVDC Dynamic Modeling Guide; Issued april, 2014, 2014.

[2] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.